

Crop-growing skyscrapers

By the year 2050, nearly 80% of the Earth’s population will live in urban centres. Applying the most conservative estimates to current demographic trends, the human population will increase by about three billion people by then. An estimated 109 hectares of new land (about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present, throughout the world, over 80% of the land that is suitable for raising crops is in use. Historically, some 15% of that has been laid waste by poor management practices. What can be done to ensure enough food for the world’s population to live on?

The concept of indoor farming is not new, since hothouse production of tomatoes and other produce has been in vogue for some time. What is new is the urgent need to scale up this technology to accommodate another three billion people. Many believe an entirely new approach to indoor farming is required, employing cutting-edge technologies. One such proposal is for the ‘Vertical Farm’. The concept is of multi-story buildings in which food crops are grown in

Question 1-7

- Complete the sentences below.
- Choose **NO MORE THAN TWO WORDS** from the passage for each answer.
- Write your answers in boxes 1-7 on your answer sheet.

Some food plants, including **1** are already grown indoors.

Some food plants, **2** including are already grown indoors.

Some food plants, including are already grown indoors **3**

Some food **4** plants, including are already grown indoors.

Some food plants, including **5** are already grown indoors.

Some food plants, including **6** are already grown indoors.

Crop-growing skyscrapers

By the year 2050, nearly 80% of the Earth’s population will live in urban centres. Applying the most conservative estimates to current demographic trends, the human population will increase by about three billion people by then. An estimated 109 hectares of new land (about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present, throughout the world, over 80% of the land that is suitable for raising crops is in use. Historically, some 15% of that has been laid waste by poor management practices. What can be done to ensure enough food for the world’s population to live on?

The concept of indoor farming is not new, since hothouse production of tomatoes and other produce has been in vogue for some time. What is new is the urgent need to scale up this technology to accommodate another three billion people. Many believe an entirely new approach to indoor farming is required, employing cutting-edge technologies. One such proposal is for the ‘Vertical Farm’. The concept is of multi-story buildings in which food crops are grown in

Question 8-13

- Do the following statements agree with the information given in Reading Passage?
- In boxes 8-13 on your answer sheet, write

TRUE if the statement agrees with the information
FALSE if the statement contradicts the information
NOT GIVEN if there is no information on this

- 8

NOT GIVEN

Methods for predicting the Earth’s pop have recently changed.
- 9

TRUE

The crops produced in vertical farms will depend on the season.
- 10

FALSE

Fertilisers will be needed for certain crops in vertical farms.

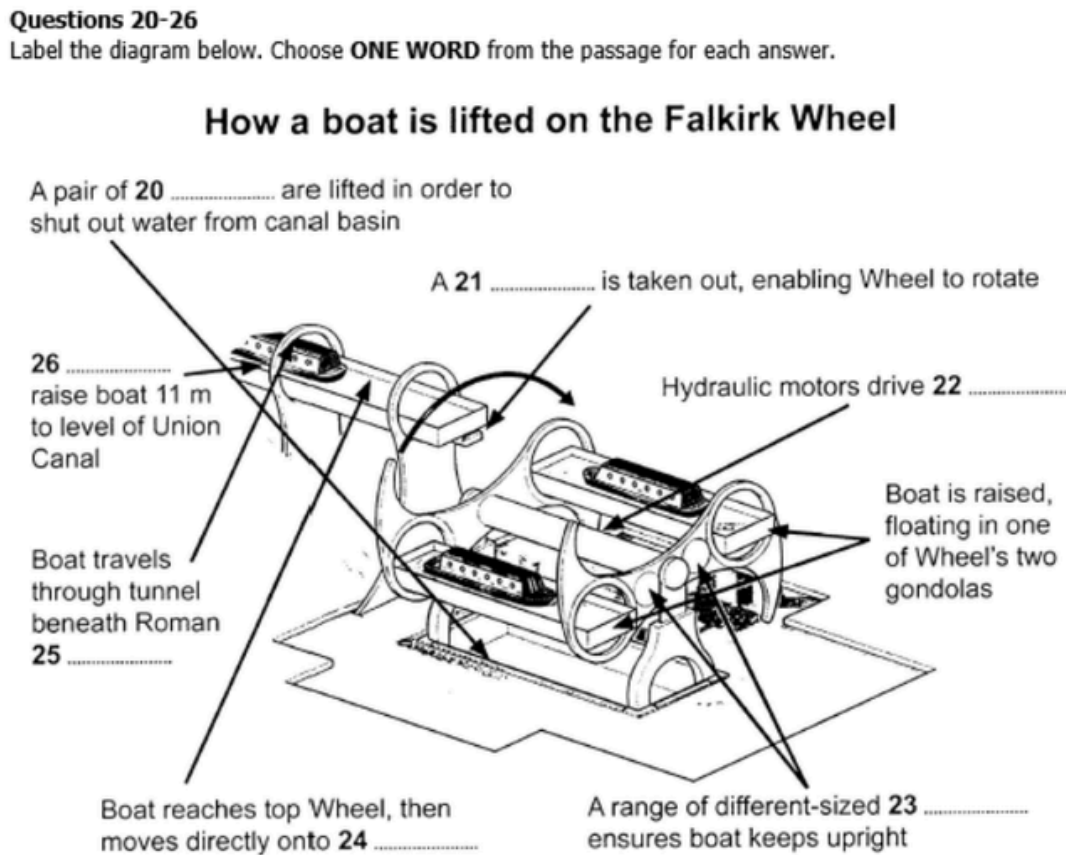
Crop-growing skyscrapers

By the year 2050, nearly 80% of the Earth’s population will live in urban centres. Applying the most conservative estimates to current demographic trends, the human population will increase by about three billion people by then. An estimated 109 hectares of new land (about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present, throughout the world, over 80% of the land that is suitable for raising crops is in use. Historically, some 15% of that has been laid waste by poor management practices. What can be done to ensure enough food for the world’s population to live on?

The concept of indoor farming is not new, since hothouse production of tomatoes and other produce has been in vogue for some time. What is new is the urgent need to scale up this technology to accommodate another three billion people. Many believe an entirely new approach to indoor farming is required, employing cutting-edge technologies. One such proposal is for the ‘Vertical Farm’. The concept is of multi-story buildings in which food crops are grown in

Question 20-26

- Label the diagram below.
- Choose **ONE WORD** from the passage for each answer.
- Write your answers in boxes 20-26 on your answer sheet.



- 20

write answer
- 21

write answer
- 22

write answer

Crop-growing skyscrapers

By the year 2050, nearly 80% of the Earth’s population will live in urban centres. Applying the most conservative estimates to current demographic trends, the human population will increase by about three billion people by then. An estimated 109 hectares of new land (about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present, throughout the world, over 80% of the land that is suitable for raising crops is in use. Historically, some 15% of that has been laid waste by poor management practices. What can be done to ensure enough food for the world’s population to live on?

The concept of indoor farming is not new, since hothouse production of tomatoes and other produce has been in vogue for some time. What is new is the urgent need to scale up this technology to accommodate another three billion people. Many believe an entirely new approach to indoor farming is required, employing cutting-edge technologies. One such proposal is for the ‘Vertical Farm’. The concept is of multi-story buildings in which food crops are grown in

Question 27-29

- Reading Passage 3 has eight paragraphs **A-H**
- Which paragraph contains the following information?
- Write the correct letter, **A-H**, in boxes **27-29** on your answer sheet.

27

A

▼

Methods for predicting the Earth’s pop have recently changed.

28

C

▼

The crops produced in vertical farms will depend on the season.

29

A

▼

B

C

D

E

F

G

H

Fertilisers will be needed for certain crops in vertical farms.

Crop-growing skyscrapers

By the year 2050, nearly 80% of the Earth’s population will live in urban centres. Applying the most conservative estimates to current demographic trends, the human population will increase by about three billion people by then. An estimated 109 hectares of new land (about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present, throughout the world, over 80% of the land that is suitable for raising crops is in use. Historically, some 15% of that has been laid waste by poor management practices. What can be done to ensure enough food for the world’s population to live on?

The concept of indoor farming is not new, since hothouse production of tomatoes and other produce has been in vogue for some time. What is new is the urgent need to scale up this technology to accommodate another three billion people. Many believe an entirely new approach to indoor farming is required, employing cutting-edge technologies. One such proposal is for the ‘Vertical Farm’. The concept is of multi-story buildings in which food crops are grown in

Question 30-36

- Complete the table below.
- Choose **ONE WORD** from the passage for each answer.
- Write your answers in boxes on your answer sheet.

GEO-ENGINEERING PROJECTS

Procedure	Aim
put a large number of tiny spacecraft into orbit far above Earth	to encourage 30 to form
release aerosol 31 sprays into the stratosphere	to encourage to 32 form
change the 33 of direction of	to encourage to form 34

Crop-growing skyscrapers

By the year 2050, nearly 80% of the Earth’s population will live in urban centres. Applying the most conservative estimates to current demographic trends, the human population will increase by about three billion people by then. An estimated 109 hectares of new land (about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present, throughout the world, over 80% of the land that is suitable for raising crops is in use. Historically, some 15% of that has been laid waste by poor management practices. What can be done to ensure enough food for the world’s population to live on?

The concept of indoor farming is not new, since hothouse production of tomatoes and other produce has been in vogue for some time. What is new is the urgent need to scale up this technology to accommodate another three billion people. Many believe an entirely new approach to indoor farming is required, employing cutting-edge technologies. One such proposal is for the ‘Vertical Farm’. The concept is of multi-story buildings in which food crops are grown in

Question 37-40

- Look at the following statements (Questions 37-40) and the list of scientists below.
- Match each statement with the correct scientist, **A-D**.
- Write the correct letter, **A-D**, in boxes **37-40** on your answer sheet.

GEO-ENGINEERING PROJECTS

A	Roger Angel
B	Phil Rasch
C	Dan Lunt
D	Martin Sommerkorn

The effects of geo-engineering may not be long-lasting. **37**

A

▼

The effects of geo-engineering may not be long-lasting. **38**

B

▼

The effects of geo-engineering may not be long-lasting. **39**

E

▼

The effects of geo-engineering may not be long-lasting. **40**

A

▼

B

C

D